

Alexander Cavoli

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EDUCATION

Rensselaer Polytechnic Institute (RPI)

B.S. Computer Science & Mathematics Dual Major; GPA: 3.92; Dean's Honors List

Troy, NY

2024 – Present

EXPERIENCE

PresentationAssistants.com — Part-Time Developer

Feb 2025 – Present

Troy, NY

- Develop features for a browser-based classroom platform under Professor Jidong Xiao, contributing to a system spanning approximately 225,000 lines of code and 494 source files used across seven undergraduate computer science courses.
- Implemented a real-time collaborative code editor enabling presenters and students to edit, compile, and execute code in-browser, along with authentication, login-attempt tracking, and administrative tools for user, project, and scheduling management.
- Contributed to synchronized classroom infrastructure supporting live code, presentations, terminal sessions, and interactive student participation across classes ranging from 20 to approximately 300 students.

Abele Technologies — Cybersecurity Specialist Intern

June 2026 – August 2026

Manchester, CT

- Supported cybersecurity and IT operations across managed client environments by developing custom security and automation tools and assisting with client-facing technical issues.
- Automated Windows disk-space remediation with a self-healing script that freed storage and generated a service ticket when usage remained above 90%. Built a safeguarded BitLocker deployment tool that enabled encryption and securely stored recovery keys in the company's password manager.

Arista Networks — CloudVision Portal Team Intern

June 2024 – Aug 2024

Remote / Santa Clara, CA

- Migrated internal CloudVision Portal tooling from the legacy `rcvpapi` wrapper to Arista's `cvprac` API, implementing missing API functionality for configlet synchronization, device operations, and automated task execution and status polling.
- Collaborated with senior developers to test and deploy production features while contributing improvements to existing development tools and workflows.

PUBLICATIONS

Presentation Assistants: A Unified Platform for Real-Time Synchronization and Learner-Controlled Navigation

ACM SIGCSE 2027

- **First author**; co-authored with Jack Hankin, Yanzhen Lu, Zeyu Chen, Zhengrui Qin, and Jidong Xiao. Peer-reviewed paper accepted to the ACM SIGCSE Technical Symposium 2027.
- Evaluated a real-time classroom platform through a two-year deployment across seven undergraduate CS courses, analyzing student engagement, participation, and interaction with synchronized instructional content.

PROJECTS

University of Delaware Hackathon Winner — TrailMate.Tech | *Node.js, JavaScript, CSS, Snowflake* Mar 2026

- Built a campus walking companion designed to help students meet new people and make walking more social, contributing across front-end UI, backend logic, and database integration.
- Integrated AR pathing with real-time guidance, building names, and distance updates; implemented live walk tracking, interactive maps, and AI-driven trend detection using Snowflake.

RPI Hackathon Winner — ByteBites.Tech | *Node.js, JavaScript, CSS*

Nov 2025

- Built a campus dining platform with allergy, meal-preference, and goal-based nutrition filtering alongside AI-generated feedback.
- Reverse engineered Sodexo's closed-source API to retrieve and process campus dining menu data for the platform.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, JavaScript, SQL, PHP, HTML/CSS

Tools & Technologies: Git, Linux, Node.js, MySQL, WebSockets, WebAssembly, REST APIs, Snowflake